LARSON—MATH 310–CLASSROOM WORKSHEET 6
Getting started with Numpy in Jupyter Notebooks.

Set up your JUPYTER notebook for this work.

1. Create a new Jupyter Notebook with Python as the kernel.

   If you don't have a preferred Jupyter Notebooks et-up, use Google Colab (`colab.google.com`), login with your VCU credentials, and use that. It's free, and part of the Google suite VCU pays for. Google Coalb has all the packages we'll need from our text including `numpy` and `matplotlib`.

2. Call this file **310-c06.ipynb**.

3. Make sure you have PYTHON as the *kernel*.

Python Basics

1. What is a `list` in Python?

2. What is *list slicing*?

3. Why are lists *mutable*?

4. What is a similar data structure that is **not** mutable?

5. What is a *list comprehension* in Python?

6. How do you code a `for loop` in Python? Write a for loop that prints the even integers from 0 to 20.

   Coding Linear Algebra with Numpy

7. Open the **Python Companion for VMLS** pdf:

   `https://ses.library.usyd.edu.au/bitstream/handle/2123/21370/vmls-python-companion.pdf` (also linked on the course web page).

   We'll talk through and code the examples in the book, annotating at least pdf pahe numbers along the way.

Getting your classwork recorded

1. Save your Jupyter Notebook file (`310-c06.ipynb` file).

2. Send me an email (`clarson@vcu.edu`) with an informative header like "Math 310 - c06 homework attached" (so that it will be properly recorded and findable when I search for these emails).

3. Remember to **attach** the `.ipynb` file for this assignment.